

Khulna University of Engineering & Technology
Department of Building Engineering & Construction Management
B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2016
Math 2123
(Mathematics - III)

Full Marks: 210

Time: 3 hrs

- N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define with example: (i) Matrix, (ii) Scalar matrix, (iii) Hermitian matrix and (iv) Trace of a matrix. (12)
- (b) If A is a square matrix, express the matrix A as a sum of a Hermitian matrix and a skew-Hermitian matrix. (10)
- (c) If $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$ and I is the unit matrix of order 3, evaluate $A^2 + 3A^T - 7A + 9I$. (13)
2. (a) Define inverse of a matrix. When does it exist? Write down the following system of equations in matrix form and find the inverse of coefficient matrix by using elementary (row) transformations. Hence solve the system using matrix method: (17)

$$\begin{aligned} x - 3y - 8z &= -10 \\ 3x + y &= -4 \\ 2x + 5y + 6z &= 13 \end{aligned}$$
- (b) Discuss different types of elementary transformations which are valid on a matrix. (05)
- (c) Find the row-reduced echelon form and hence the normal form of the matrix $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$. What is its rank? (13)
3. (a) For what value (s) of λ , the system of equations, (15)

$$\begin{aligned} 3x - y + 4z &= 3 \\ x + 2y - 3z &= -2 \\ 6x + 5y + \lambda z &= -3 \end{aligned}$$

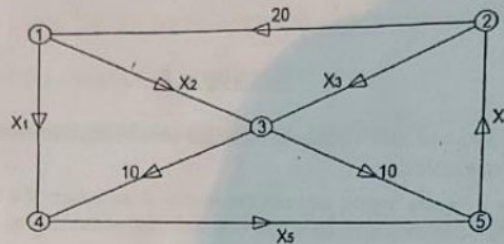
Possesses (i) a unique solution? and (ii) infinitely many solutions?
Determine the solution for case (ii).
- (b) Why a homogenous system is always consistent? What do we mean by trivial and non-trivial solution of a homogeneous system? Solve the system: (15)

$$\begin{aligned} 4x_1 - x_2 + 2x_3 + x_4 &= 0 \\ 2x_1 + 3x_2 - x_3 - 2x_4 &= 0 \\ 7x_2 - 4x_3 - 5x_4 &= 0 \\ 2x_1 - 11x_2 + 7x_3 + 8x_4 &= 0 \end{aligned}$$

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- (c) Define orthogonal matrix. Is the matrix $\frac{1}{3} \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}$ orthogonal? (12)

4. (a) Set up a system of linear equations to represent the network shown in the figure: (12)



What type of solution does the system possess?

- (b) Define eigen values and eigen vectors. Determine the Eigen values and the corresponding eigen vectors of the matrix. (18)

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

- (c) If a non-singular matrix A is symmetric show that A^{-1} is also symmetric. (05)

Section B

5. (a) Prove that $\underline{A} \times (\underline{B} \times \underline{C}) = \underline{B}(\underline{A} \cdot \underline{C}) - \underline{C}(\underline{A} \cdot \underline{B})$ for any three vectors $\underline{A}$, $\underline{B}$ and $\underline{C}$. (12)
Using this result, show that $\underline{A} \times (\underline{B} \times \underline{C}) + \underline{B} \times (\underline{C} \times \underline{A}) + \underline{C} \times (\underline{A} \times \underline{B}) = 0$.
- (b) The diagonal of a parallelogram are given by the vectors $\underline{A} = 3\hat{i} - 4\hat{j} - \hat{k}$ and $\underline{B} = 2\hat{i} + 3\hat{j} - 6\hat{k}$. Prove that the parallelogram is a rhombus and determine the length of its sides. (12)
- (c) Determine whether the vectors are linearly independent or linearly dependent: (11)
 $\underline{A} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\underline{B} = \hat{i} - 4\hat{k}$, $\underline{C} = 4\hat{i} + 3\hat{j} - \hat{k}$.
6. (a) Find equations for the tangent plane and normal line to the surface $z = x^2 + y^2$ at the point $(2, -1, 5)$. (12)
- (b) In what direction from the point $(2, 1, -1)$ in the directional derivative of $\varphi = x^2 y z^3$ a maximum? What is the magnitude of this maximum? (12)
- (c) Evaluate (i) $\nabla^2(\ln r)$ (ii) $\nabla f(r)$. (11)
7. (a) The acceleration $\underline{a}$ of an object at any time t is given by $\underline{a} = -g\hat{j}$, where g is a constant. At $t = 0$ the velocity is given by $\underline{V} = V_0 \cos \theta_0 \hat{i} + V_0 \sin \theta_0 \hat{j}$ and the displacement $\underline{r} = \underline{0}$. Find $\underline{V}$ and $\underline{r}$ at any time $t > 0$. (12)

(b) If $\underline{A} = (2y+3)\underline{i} + xz\underline{j} + (yz-x)\underline{k}$, evaluate $\int_C \underline{A} \cdot d\underline{r}$ along the following paths C.

(i) the straight lines from $(0,0,0)$ to $(0,0,1)$, then to $(0,1,1)$ and then to $(2,1,1)$.

(ii) the straight line joining $(0,0,0)$ and $(2,1,1)$.

(c) Evaluate $\iiint_V (2x+y) dV$, where V is the closed region bounded by the (11)

cylinder $z = 4 - x^2$ and the planes $x=0$, $y=0$, $y=2$ and $z=0$.

8. (a) Find Laplace transforms of (i) $F''(t)$ (ii) $e^{4t} + 4t^3 - 2\sin 3t + 4\cos 5t$. (12)

(b) Evaluate $L^{-1}\left\{\frac{1}{s^3(s^2+4)}\right\}$. (11)

(c) Solve the differential equation using the Laplace transform: $Y'' + a^2Y = F(t)$; (12)

$Y(0) = 1$, $Y'(0) = -2$.

$$L^{-1} \frac{1}{s^3(s^2+4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{Ds+E}{s^2+4}$$

$$1 = A(s^2+4s^2) + B(s^2+4) + C(s^2+4) + (Ds+E)s^3$$

$$\Rightarrow As^4 + 4As^2 + Bs^3 + 4Bs + Cs^2 + 4C + Ds^3 + Es^3$$

$$A+D=0 \quad \text{--- (i)}$$

$$B+E=0 \quad \text{--- (ii)}$$

$$4A+C=0 \quad \text{--- (iii)}$$

$$4B=0 \Rightarrow B=0$$

$$4C=1 \Rightarrow C=\frac{1}{4}$$

$$\therefore E=0$$

$$4A=\frac{1}{4} \Rightarrow A=\frac{1}{16}$$

$$D=\frac{1}{16}$$

$$\therefore \frac{1}{16} \left(\frac{1}{s} + \frac{1}{s^3} + \frac{1}{s^2+4} \right)$$

$$= \frac{1}{16} + \frac{1}{4} \frac{t^2}{2!} + \frac{1}{16} \cos 2t$$

$$= -\frac{1}{16} + \frac{1}{8} t^2 + \frac{1}{16} \cos 2t$$